

# FACTORY MANAGEMENT SCHEDULER

A Java-Based Command-Line Application for Factory Production Scheduling

## PROJECT REPORT

|                         |                       |
|-------------------------|-----------------------|
| <b>Student Name</b>     | <b>Karra Priyanka</b> |
| Registration / Roll No. | 25BAI11430            |
| Course / Subject        | VITyarthi Project     |
| Programming Language    | Java                  |
| University              | VIT Bhopal University |
| Submission Date         | 18 September 2026     |

Submitted for academic evaluation

## TABLE OF CONTENTS

1. Project Title
2. Abstract
3. Problem Statement
4. Objectives
5. Main Modules
6. System Flowchart
7. Job Management Flowchart
8. Machine Management Flowchart
9. Scheduling Flowchart
10. Scheduling Algorithm
11. System Architecture
12. Data Structures Used
13. Input → Processing → Output
14. Input Information
15. Output Information
16. Use Case Diagram
17. Functional Requirements
18. Non-Functional Requirements
19. Java Concepts Used
20. Project Workflow
21. Test Cases
22. Actual Execution Results
23. Advantages
24. Limitations
25. Future Enhancements
26. Project Structure
27. Compilation and Execution
28. Overall System Diagram
29. Conclusion
- Appendix

# 1. PROJECT TITLE

## Factory Management Scheduler

A Java-based command-line application for organizing production jobs, managing machine availability, and generating a simple priority-based factory schedule.

Author: Karra Priyanka

Registration / Roll No.: 25BAI11430

Course: VITyarthi Project

Programming Language: Java

VIT Bhopal University

# 2. ABSTRACT

The Factory Management Scheduler is a Java-based command-line application developed to organize basic factory production activities. The system allows users to manage production jobs and factory machines and generate a simple production schedule. Users can add, view, and delete jobs using job name, duration, and priority. Machines can be added and their availability status can be managed.

The scheduling module arranges jobs according to priority and assigns them to available machines while calculating starting and ending times. The implementation uses standard Java features such as ArrayList, Scanner, methods, loops, conditional statements, sorting, and arithmetic operations.

# 3. PROBLEM STATEMENT

Factories may have several production jobs that need to be completed using available machines. Managing these jobs manually can make it difficult to keep track of job priorities, processing duration, and machine availability.

The project provides a simple computerized system where the user can enter production jobs and machines and generate an organized schedule based on job priority and machine availability.

# 4. OBJECTIVES

- Manage production jobs and store job names and processing durations.
- Assign priorities to jobs.
- Add and manage factory machines.
- Track machine availability.
- Arrange jobs according to priority.
- Assign available machines to jobs.
- Calculate job starting and ending times.
- Provide the schedule through the command line.
- Demonstrate practical Java programming concepts.

# 5. MAIN MODULES

|                              |
|------------------------------|
| FACTORY MANAGEMENT SCHEDULER |
| ↓                            |



|                                                                                            |
|--------------------------------------------------------------------------------------------|
| JOB MANAGEMENT: Add / View / Delete Jobs                                                   |
| ↓                                                                                          |
| MACHINE MANAGEMENT: Add / View / Change Status                                             |
| ↓                                                                                          |
| SCHEDULING: Sort by Priority → Find Available Machine → Calculate Time → Generate Schedule |

## 6. SYSTEM FLOWCHART

|                                                         |
|---------------------------------------------------------|
| START                                                   |
| ↓                                                       |
| Display Main Menu                                       |
| ↓                                                       |
| Enter Choice                                            |
| ↓                                                       |
| Job Management / Machine Management / Scheduling / Exit |
| ↓                                                       |
| Perform Selected Operation                              |
| ↓                                                       |
| Return to Menu or Exit                                  |
| ↓                                                       |
| END                                                     |

## 7. JOB MANAGEMENT FLOWCHART

|                                            |
|--------------------------------------------|
| JOB MANAGEMENT                             |
| ↓                                          |
| Display Menu                               |
| ↓                                          |
| Add Job: Name → Duration → Priority → Save |
| ↓                                          |
| View Jobs: Check → Display                 |
| ↓                                          |
| Delete Job: Select → Delete → Confirm      |

|              |
|--------------|
| ↓            |
| Back to Menu |

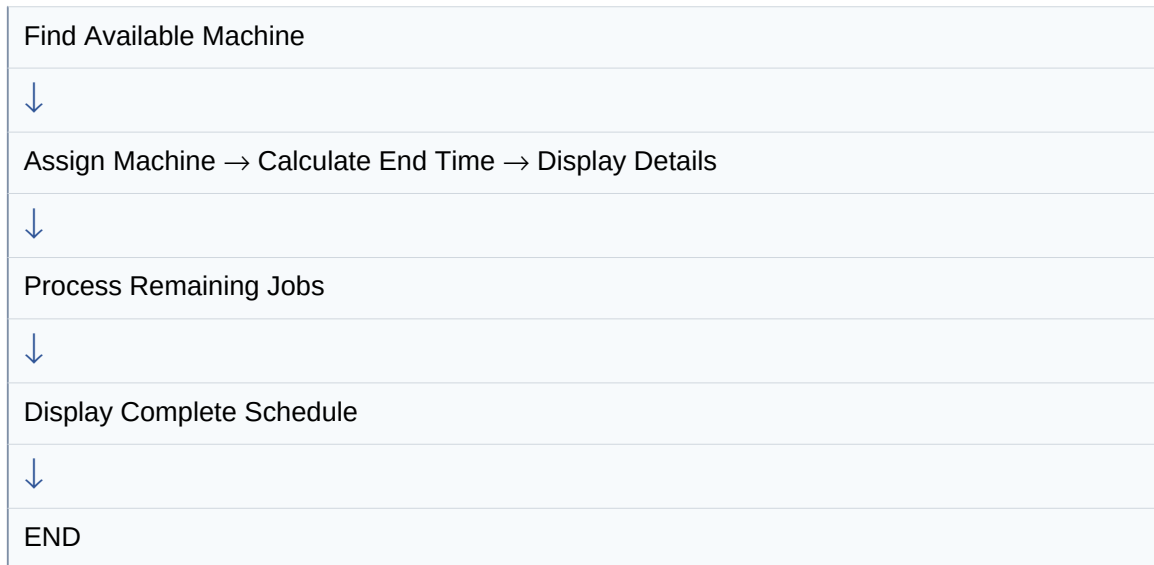
Job priority is defined as 1 = High, 2 = Medium, and 3 = Low.

## 8. MACHINE MANAGEMENT FLOWCHART

|                                             |
|---------------------------------------------|
| MACHINE MANAGEMENT                          |
| ↓                                           |
| Display Menu                                |
| ↓                                           |
| Add Machine / View Machines / Change Status |
| ↓                                           |
| Add: Enter Name → Set Available → Save      |
| ↓                                           |
| Change Status: Available ↔ Busy             |
| ↓                                           |
| Back to Menu                                |

## 9. SCHEDULING FLOWCHART

|                                  |
|----------------------------------|
| START                            |
| ↓                                |
| Check Jobs                       |
| ↓                                |
| If no jobs → display message     |
| ↓                                |
| Check Machines                   |
| ↓                                |
| If no machines → display message |
| ↓                                |
| Sort Jobs by Priority: 1 → 2 → 3 |
| ↓                                |
| Set Time = 8:00 AM               |
| ↓                                |



## 10. SCHEDULING ALGORITHM

**Step 1:** Check whether jobs are available.

**Step 2:** Check whether machines are available.

**Step 3:** Create an order containing all job positions.

**Step 4:** Sort jobs according to priority: 1 → High, 2 → Medium, 3 → Low.

**Step 5:** Set the initial working time to 8:00 AM.

**Step 6:** Search for an available machine.

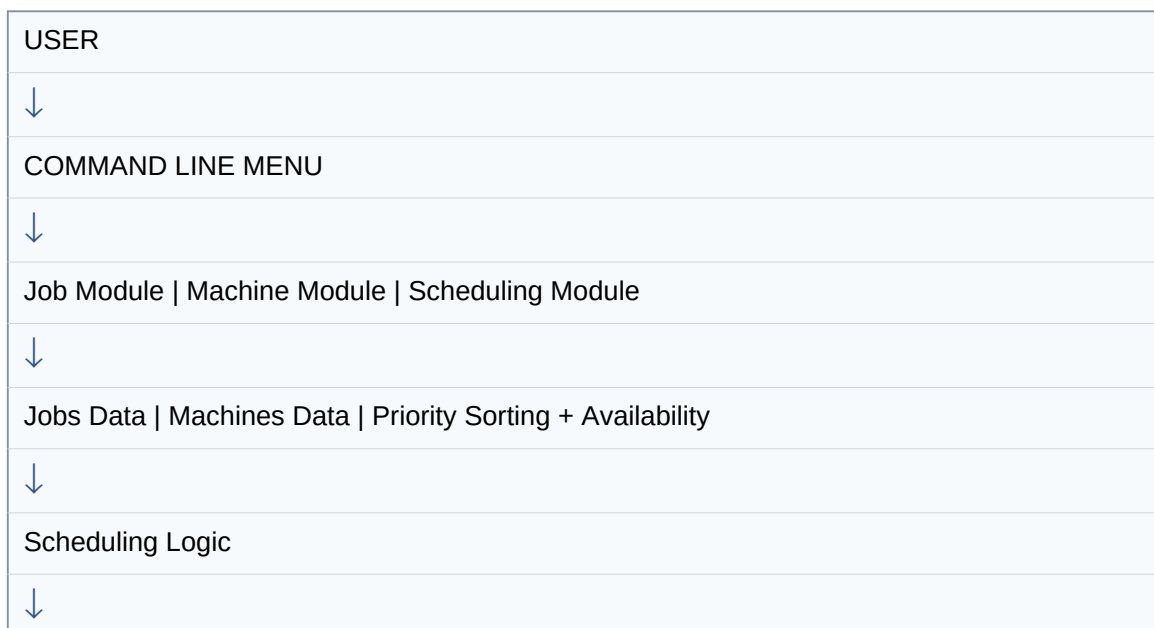
**Step 7:** Assign the selected machine to the current job.

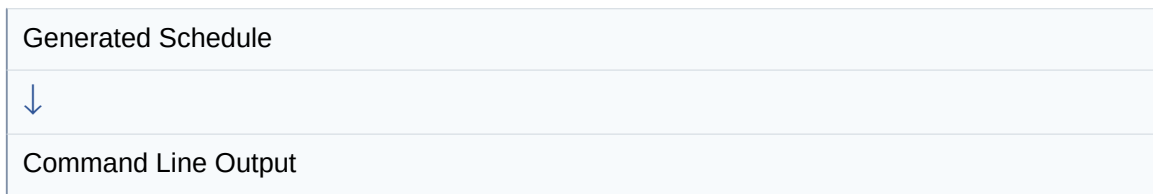
**Step 8:** Calculate End Time = Start Time + Job Duration.

**Step 9:** Display Job Name, Machine Name, Start Time, End Time, and Priority.

**Step 10:** Continue until all jobs are processed.

## 11. SYSTEM ARCHITECTURE





## 12. DATA STRUCTURES USED

The project uses Java ArrayList collections to store related information dynamically during program execution.

| Data Structure | Purpose                            |
|----------------|------------------------------------|
| jobs           | Stores job names                   |
| durations      | Stores job duration                |
| priorities     | Stores job priority                |
| machines       | Stores machine names               |
| machineStatus  | Stores machine availability status |

Related values use the same index. For example, index 0 can represent Gear Production, duration 3 hours, priority 1.

## 13. INPUT → PROCESSING → OUTPUT



## 14. INPUT INFORMATION

| Category      | Fields                       | Example                              |
|---------------|------------------------------|--------------------------------------|
| Job Input     | Job Name, Duration, Priority | Gear Production, 3 hours, Priority 1 |
| Machine Input | Machine Name, Status         | CNC Machine, Available               |

## 15. OUTPUT INFORMATION

The scheduler is designed to display Job, Machine, Starting Time, Ending Time, and Priority.

```

Job: Gear Production
Machine: CNC Machine
Time: 8:00 - 11:00
Priority: 1

```

## 16. USE CASE DIAGRAM

|                                                                         |
|-------------------------------------------------------------------------|
| USER                                                                    |
| ↓                                                                       |
| Manage Jobs: Add / View / Delete                                        |
| ↓                                                                       |
| Manage Machines: Add / View / Change Status                             |
| ↓                                                                       |
| Generate Schedule: Sort / Find Machine / Calculate Time / View Schedule |
| ↓                                                                       |
| Exit                                                                    |

## 17. FUNCTIONAL REQUIREMENTS

| ID   | Requirement                             |
|------|-----------------------------------------|
| FR1  | User can add a job                      |
| FR2  | User can view jobs                      |
| FR3  | User can delete a job                   |
| FR4  | User can add a machine                  |
| FR5  | User can view machines                  |
| FR6  | User can change machine status          |
| FR7  | System sorts jobs according to priority |
| FR8  | System assigns available machines       |
| FR9  | System calculates schedule time         |
| FR10 | System displays generated schedule      |

## 18. NON-FUNCTIONAL REQUIREMENTS

| Category        | Description                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Usability       | Simple command-line interface.                                                |
| Performance     | Processes entered jobs and machines quickly for a small scheduling scenario.  |
| Reliability     | Checks whether jobs and machines are available before scheduling.             |
| Portability     | Runs on systems with a compatible Java JDK.                                   |
| Maintainability | Uses separate methods for job management, machine management, and scheduling. |

## 19. JAVA CONCEPTS USED

| Concept            | Use                                                       |
|--------------------|-----------------------------------------------------------|
| Scanner            | Receives user input.                                      |
| ArrayList          | Stores jobs, machines, durations, priorities, and status. |
| Methods            | Separates major program functions.                        |
| Loops              | Processes multiple jobs and machines.                     |
| if-else            | Processes menu choices and availability conditions.       |
| Boolean            | Represents machine availability.                          |
| Arithmetic         | Supports time calculations.                               |
| Sorting / Lambda   | Arranges jobs according to priority.                      |
| Modular Arithmetic | Included in the implementation concepts.                  |

## 20. PROJECT WORKFLOW



END

## 21. TEST CASES

| Test | Action            | Expected Result                 |
|------|-------------------|---------------------------------|
| TC01 | Add new job       | Job added successfully          |
| TC02 | View jobs         | Stored jobs displayed           |
| TC03 | Delete valid job  | Selected job deleted            |
| TC04 | Add machine       | Machine added as Available      |
| TC05 | View machines     | Machines and status displayed   |
| TC06 | Change status     | Status toggles Available / Busy |
| TC07 | Generate schedule | Schedule generated              |
| TC08 | No jobs           | No jobs available message       |
| TC09 | No machines       | No machines available message   |
| TC10 | Invalid choice    | Invalid choice message          |

## 22. ACTUAL EXECUTION RESULTS

The following five screenshots are the actual images supplied with the project. They have been inserted according to the operation visibly demonstrated in each screenshot. No unverified scheduling screenshot has been created.

### Result 1 — Compilation, Execution and Machine Addition

```
C:\Users\D Jai Deepak\OneDrive\Attachments\Documents\New Folder\FactoryManagementScheduler>javac Main.java
C:\Users\D Jai Deepak\OneDrive\Attachments\Documents\New Folder\FactoryManagementScheduler>java Main

=== FACTORY MANAGEMENT SCHEDULER ===
1. Job Management
2. Machine Management
3. Scheduling
4. Exit
Enter choice: 2

--- MACHINE MANAGEMENT ---
1. Add Machine
2. View Machines
3. Change Machine Status
4. Back
Enter choice: 1
Enter machine name: CNC Machine
Machine added successfully.
```

Figure 1: Java compilation/execution and entry of CNC Machine.

The screenshot shows **javac Main.java** followed by **java Main**. The program enters Machine Management and accepts the machine name CNC Machine, demonstrating successful launch and machine input.

### Result 2 — Machine Management and Machine Viewing

```
--- MACHINE MANAGEMENT ---
1. Add Machine
2. View Machines
3. Change Machine Status
4. Back
Enter choice: 4

=== FACTORY MANAGEMENT SCHEDULER ===
1. Job Management
2. Machine Management
3. Scheduling
4. Exit
Enter choice: 2

--- MACHINE MANAGEMENT ---
1. Add Machine
2. View Machines
3. Change Machine Status
4. Back
Enter choice: 2
1. CNC Machine | Status: Available
2. Lathe Machine | Status: Available
```

Figure 2: Machine Management menu and stored machine availability.

The View Machines operation displays **CNC Machine** and **Lathe Machine**, both with status **Available**.

### Result 3 — Job Management: Add Job

```
--- JOB MANAGEMENT ---
1. Add Job
2. View Jobs
3. Delete Job
4. Back
Enter choice: 1
Enter job name: hod
Enter duration in hours: 123
Enter priority (1-High, 2-Medium, 3-Low): 2
Job added successfully.

--- JOB MANAGEMENT ---
1. Add Job
2. View Jobs
3. Delete Job
4. Back
Enter choice: |
```

Figure 3: Adding a production job.

The screenshot records job name **hod**, duration **123 hours**, priority **2 (Medium)**, followed by **Job added successfully**.

#### Result 4 — Job Management: Delete Job

```
--- JOB MANAGEMENT ---
1. Add Job
2. View Jobs
3. Delete Job
4. Back
Enter choice: 3
1. hod | Duration: 123 hrs | Priority: 2
Enter job number to delete: 1
Job deleted successfully.

--- JOB MANAGEMENT ---
1. Add Job
2. View Jobs
3. Delete Job
4. Back
Enter choice: |
```

Figure 4: Deleting a production job.

The stored job **hod** is selected as job number 1 and the application displays **Job deleted successfully**.

#### Result 5 — Main Menu and Empty-Job Validation

```

C:\Users\D Jai Deepak\OneDrive\Attachments\Documents\New Folder\FactoryManag
ementScheduler>javac Main.java

C:\Users\D Jai Deepak\OneDrive\Attachments\Documents\New Folder\FactoryManag
ementScheduler>java Main

=== FACTORY MANAGEMENT SCHEDULER ===
1. Job Management
2. Machine Management
3. Scheduling
4. Exit
Enter choice: 1

--- JOB MANAGEMENT ---
1. Add Job
2. View Jobs
3. Delete Job
4. Back
Enter choice: 3
No jobs available.

```

Figure 5: Main menu, Job Management, and empty-job validation.

The screenshot shows the application being launched, Job Management being selected, and the View Jobs operation returning **No jobs available**.

### Observed Execution Summary

| Feature              | Observed Evidence                                  |
|----------------------|----------------------------------------------------|
| Compilation / launch | javac Main.java and java Main visible.             |
| Machine creation     | CNC Machine entered.                               |
| Machine viewing      | CNC Machine and Lathe Machine listed as Available. |
| Job creation         | hod, 123 hours, priority 2 added successfully.     |
| Job deletion         | hod selected and deleted successfully.             |
| Empty-job validation | No jobs available message displayed.               |

### Scheduling Screenshot Note

The supplied five screenshots do not show the final generated factory schedule. The report therefore describes the scheduling functionality from the documented design, but does not claim that a schedule screenshot was observed.

## 23. ADVANTAGES

- Simple and easy to operate.
- Uses a command-line interface.
- Provides basic job organization.
- Tracks machine availability.
- Prioritizes important jobs.
- Calculates scheduling times.
- Uses standard Java without external libraries.

- Suitable for demonstrating Java programming concepts.

## 24. LIMITATIONS

- Data is stored only while the program is running.
- No database is used.
- No graphical user interface is included.
- Scheduling is based mainly on priority and machine availability.
- No real-time machine monitoring.
- Does not include complex industrial constraints.
- Generated schedules are not permanently saved.

## 25. FUTURE ENHANCEMENTS

|                              |
|------------------------------|
| CURRENT SYSTEM               |
| ↓                            |
| Database Integration         |
| ↓                            |
| GUI / Web Interface          |
| ↓                            |
| Automatic Machine Allocation |
| ↓                            |
| Production Deadlines         |
| ↓                            |
| Machine Maintenance Tracking |
| ↓                            |
| Reports and Analytics        |
| ↓                            |
| Real-Time Factory Monitoring |

- MySQL or database storage.
- Graphical user interface.
- Web-based dashboard.
- Automatic machine allocation.
- Production deadlines.
- Machine maintenance records.
- Employee management.

- Production reports.
- Schedule history.
- Real-time machine monitoring.

These are proposed future enhancements and are not presented as current features.

## 26. PROJECT STRUCTURE

```
FactoryManagementScheduler /  
■  
■■■ Main.java  
■  
■■■ README.md
```

Main.java contains the main program, main menu, job management, machine management, scheduling, and related viewing operations.

## 27. COMPILATION AND EXECUTION

|                                       |
|---------------------------------------|
| Open Terminal / Command Prompt        |
| ↓                                     |
| Navigate to Project Folder            |
| ↓                                     |
| Compile: <code>javac Main.java</code> |
| ↓                                     |
| Run: <code>java Main</code>           |
| ↓                                     |
| Main Menu Appears                     |

```
cd FactoryManagementScheduler  
javac Main.java  
java Main
```

## 28. OVERALL SYSTEM DIAGRAM

|                                                   |
|---------------------------------------------------|
| FACTORY MANAGEMENT SCHEDULER                      |
| ↓                                                 |
| JOB MANAGEMENT: Add / View / Delete               |
| ↓                                                 |
| MACHINE MANAGEMENT: Add / View / Change Status    |
| ↓                                                 |
| SCHEDULING: Priority Sorting + Machine Assignment |
| ↓                                                 |
| TIME CALCULATION                                  |

|                    |
|--------------------|
| ↓                  |
| GENERATED SCHEDULE |
| ↓                  |
| USER OUTPUT        |

## 29. CONCLUSION

The Factory Management Scheduler demonstrates how Java programming can be applied to a practical production management problem. The application provides three main functions: job management, machine management, and scheduling. Users can maintain production jobs, manage machine availability, and generate a priority-based schedule with calculated start and end times.

The project uses fundamental Java concepts such as ArrayList, Scanner, methods, loops, conditional statements, sorting, and arithmetic operations. The supplied execution screenshots provide evidence of program launch, machine management, job addition, job deletion, and empty-job validation. The final scheduling output is part of the designed functionality, but a corresponding screenshot was not included in the five supplied images.

Although the current system is a simple command-line application, it provides a foundation for database storage, graphical interfaces, machine maintenance tracking, deadlines, reporting, and more advanced scheduling features.

## APPENDIX A — SOURCE CODE STRUCTURE

```
import java.util.*;

public class Main {
    static ArrayList<String> jobs = new ArrayList<>();
    static ArrayList<Integer> durations = new ArrayList<>();
    static ArrayList<Integer> priorities = new ArrayList<>();
    static ArrayList<String> machines = new ArrayList<>();
    static ArrayList<Boolean> machineStatus = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        while (true) {
            // Main menu
            // 1. Job Management
            // 2. Machine Management
            // 3. Scheduling
            // 4. Exit
        }
    }

    // jobManagement()
    // machineManagement()
    // scheduling()
    // viewJobs()
}
```

## APPENDIX B — FIGURE INDEX

| Figure   | Description                                  |
|----------|----------------------------------------------|
| Figure 1 | Compilation / execution and machine addition |
| Figure 2 | Machine Management and machine availability  |

| Figure   | Description                        |
|----------|------------------------------------|
| Figure 3 | Job Management — Add Job           |
| Figure 4 | Job Management — Delete Job        |
| Figure 5 | Main Menu and empty-job validation |